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Evaluation of Virulence, Antimicrobial Resistance and Biofilm Forming Potential of Methicillin-Resistant *Staphylococcus aureus* (MRSA) Isolates from Bovine Suspected with Mastitis

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Abstract

Methicillin-resistant *Staphylococcus aureus* (MRSA) is a pathogen that poses a significant threat in cases of chronic mastitis in dairy animals. The ability of MRSA to persist in the host is attributed to various virulence factors, genes encoding surface adhesins, and determinants of antibiotic resistance, which provide it a survival advantage. This investigation focused to determine the virulence factors, antimicrobial resistance (AMR) profile and biofilm production potential of 46 MRSA isolates from 300 bovine mastitis milk samples. The AMR profile revealed a high level of resistance, with 46 and 42 isolates resistant to cefoxitin and oxacillin, respectively, followed by 24 and 12 isolates resistant to lomefloxacin and erythromycin, respectively. Only 2 isolates resistant to tetracycline and none were resistant to chloramphenicol. The study also evaluated various virulence factors such as *coa* (n = 46), *nuc* (n = 35) *hlg* (n = 36), *pvl* (n = 14), *tsst-1*(n = 28) *spa* (n = 39) and enterotoxin genes *sea* (n = 12) and *seg* (n = 28) and identified antibiotic resistance determinants *mecA* and *blaZ* in 46 and 27 isolates, respectively. Intercellular adhesion genes *icaA* and *icaD* were present in 40 and 43 isolates, respectively and surface adhesion genes *ebps*, *fnbpA*, *eno*, *sasG*, *cna*, and *bap* were found in 43, 40, 38, 26, 21 and 1 isolates, respectively. Microtiter plate (MTP) assay revealed that 29 MRSA isolates were capable of producing biofilms, whereas 17 were not. Biofilms producing MRSA isolates possessed adhesion genes, virulence factors, toxin genes and AMR genes that may act synergistically towards a chronic disease progression, illness and severe damage to the udder, which generally last for several months and very challenging to cure.

Introduction

Bovine mastitis is a prevalent inflammatory disease affecting the mammary gland in dairy cattle and buffalo, causing a significant impact on milk quality and quantity, premature culling, treatment expenses, and additional labor costs, leading to substantial economic losses for the dairy industry and its stakeholders [1]. In India, the annual economic loss due to both Sub-clinical mastitis (SCM) and Clinical mastitis (CM) is estimated to be US$ 98,228 million [2]. Mastitis is characterized into three types, namely sub-clinical, clinical and chronic, with SCM being the major source of financial losses due to its complex nature [2]. CM is a persistent inflammation of the mammary gland characterized by udder enlargement, milk with flakes and clots, and watery milk, although it is rare. In India, Methicillin-resistant *Staphylococcus aureus* (MRSA) is one of the leading causes of bovine mastitis, with *S. aureus* infection reported in 10–44% of mastitis cases, resulting in significant financial losses for dairy farms.[3, 4].

S. aureus is known for its pathogenicity, which is associated with the acquisition of multiple factors that confer resistance to various drugs. These factors include enterotoxins, leukocidins, haemolysins, superantigens, immune evasive surface factors, capsular polysaccharides, and the robust ability of biofilm production. Among these factors, Panton Valentine leukocidin (PVL) and haemolysin (Hla and Hlb) are capable of forming pores in host neutrophils, macrophages and endothelial cells, leading to leukocytolysis [5]. Furthermore, toxic shock syndrome toxin-1 (Tsst-1) is

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known to act as a superantigen, triggering T-cell responses with massive cytokine production, which can result in toxic shock syndrome [3]. The relationship between virulence and antibiotic resistance in *S. aureus* is complex, with previous studies suggesting that genetic elements that confer both antibiotic resistance and virulence are intertwined. Communication between bacteria through quorum sensing and the two-component system of biofilm can lead to global changes in gene expression, enhancing virulence and promoting the acquisition of antibiotic resistance [6, 7].

MRSA is a pathogen that has been linked to various infections in both humans and animals. It is commonly found in healthcare and community settings and is a significant cause of endocarditis, bacteremia, skin and soft tissue infections in humans, as well as mastitis, udder impetigo, wound infections, tick pyaemia, and bumblefoot in animals and birds [5, 8]. Staphylococcal foodborne disease (SFD), a prevalent foodborne infection, poses a significant threat to global public health [9]. The emergence of MRSA in farm animals is a particular concern because it can spread to humans through animal-derived foods and cause resistant infections. *S. aureus* is capable of forming biofilms on both biotic and abiotic surfaces throughout the food production chain. The Staphylococcal classical enterotoxins (SEA-SEE) are responsible for 90% of staphylococcal food poisoning (SFP) outbreaks and pose a serious threat to human health [3, 9]. Apart from causing SFD in humans, in clinical settings, *S. aureus* is the most common cause of biomaterial-centered infections often linked to various implantable devices, such as central venous catheters, endotracheal tubes, intrauterine devices, mechanical heart valves (MHV), pacemakers, and contact lenses. The presence of certain mobile genetic elements (MGE), such as Tn4451/Tn4453 and TnAs3, is more common in both humans and food animals (swine and chicken) because they share them. TnpX, a member of a group of proteins called resolvase site-specific recombinases, is present in Tn4451/Tn4453 and helps insert circular molecules temporarily, indicating the potential for zoonotic transmission of antibiotic resistance genes between humans and food animals. [10].

A complex structure known as a "bacterial biofilm" is created by microorganisms that attach themselves to a surface and become covered by an extracellular matrix made up of polysaccharides, proteins, and extracellular DNA (eDNA). These biofilms create self-sustaining ecosystems [11]. Within a biofilm, cells are known to be up to 1000 times more resistant to antibiotics and antimicrobial peptides compared to planktonic cells [12]. The initial attachment of bacteria to a biotic surface is facilitated by microbial surface components recognizing adhesive matrix molecules (MSCRAMMs) that bind to host matrix proteins such as fibronectin-binding proteins FnBPA and FnBPB, clumping factor A (ClfA) and clumping factor B (ClfB), serine-aspartate repeat family proteins (SdrC, SdrD, and SdrE) and bone sialoprotein-binding (Bpb). Binding to abiotic surfaces is facilitated through hydrophobic and electrostatic interactions [13]. In the next stage, the bacteria produce exopolysaccharide (EPS) that form biofilm matrix. Staphylococcal biofilms are classified into two categories based on their EPS composition: polysaccharide-dependent and proteinaceous-dependent biofilms. The main component in polysaccharide-dependent biofilms is poly-β(1-6)-N-acetylglucosamine (PNAG),also known as polysaccharide intercellular adhesin (PIA) [13]. The production of PIA is governed by the intracellular adhesion (ica) operon which is made up of four genes: *icaA*, *icaD*, *icaB*, and *icaC*. Proteinaceous biofilms are mainly observed in MRSA isolates, whereas methicillin-sensitive *S. aureus* (MSSA) isolates are more often *ica* dependent. A third type of staphylococcal biofilm, is mainly dependent on eDNA that is released by lytic enzyme autolysin in *S. aureus*. [13].

Molecular analysis have demonstrated that widespread multidrug resistance among bovine pathogens is result of pre-existing resistance factors located on MGE, which are acquired from resistant bacteria in dairy farms and then amplified through selection [5]. Resistance in MRSA is usually conferred by horizontal transfer of MGE known as staphylococcal cassette chromosome *mec* (SCCmec), which carries a *mecA* gene encoding a penicillin-binding protein (PBP2a) with reduced affinity for methicillin and other beta-lactam antibiotics [14]. To combat bovine mastitis and human infection, antimicrobial treatment is considered the primary and most effective control measure. However, the extensive use of antibiotics in human, dairy animal, and industrial applications has led to a global increase in antibiotic-resistant bacteria (ARBs) [15]. Recent research, which utilized predicted statistical models, has estimated that there were 1.27 million deaths caused by AMR in 2019 [16].

The One-Health concept embodies an integrated approach to safeguarding the health and well-being of humans, animals, and the environment. Notably, microbes that cause mastitis have demonstrated the capacity to traverse hierarchical barriers and cause zoonotic transmissions from bovines to humans through direct contact or the consumption of raw milk or undercooked meat [17]. It is therefore imperative to effectively monitor and manage MRSA in bovine mastitis, given the significant threats it poses to humans, animals, and the environment. In this regard, the present study aims to comprehensively characterize the molecular and phenotypic characteristics of MRSA isolates from bovine mastitis, with particular emphasis on virulence factors, toxin genes, antimicrobial resistance profiles, and biofilm-forming capacity. Based on the findings of our investigation, it is reasonable to infer that the development of biofilms, coupled with various virulence factors and antibiotic resistance, could exacerbate the recurrence of difficult-to-treat and cure udder inflammation in cattle.

Materials and Methods

Ethical Approval

A total of 300 samples were collected from LUVAS Uchani, Karnal, Haryana, India under the SERB funded project entitled "**Mastitis related antibiotic resistance pattern mapping in three districts of Haryana (FILE NO. EMR/2017/004602)**" were discussed in relation to the research objectives and methodology. The protocols, materials and methods to be followed in the study and project were approved by the Institutional Biosafety Committee (IBSC), National Dairy Research Institute (NDRI), Karnal, India.

Collection of Milk Samples and Physical Examination of Mastitis

The current investigation obtained 300 milk samples from cattle and buffaloes suspected with mastitis, collected by professional veterinarians from the Regional Centre, LUVAS (Lala Lajpat Rai University of Veterinary and Animal Sciences), Uchani, Karnal, Haryana, India, between August 2021 and February 2022. The study considered three types of mastitis cases for sample collection: (a) acute cases with high fever and temperature, (b) chronic cases refractory to treatment of the mammary gland with no increase in body temperature, and (c) samples from clinical diagnostic labs for AST testing. Prior to milk sample collection, each animal underwent clinical examination to assess the presence of clinical signs associated with udder and teats, such as fibrosis, inflammation, pain, visible injury or lesion, atrophy of the tissue, pathological conditions like heat, hardness, and redness, as well as changes in milk parameters, such as colour, appearance of flakes, clots, or pus in the milk. Milk samples were collected from mastitis-suspected animals, characterized based on physical appearance, including blood-tinged milk (BTM), straw-coloured milk (SCM), watery milk (WM), and milk with pus and tissue (PTM), and provided with collection and sample codes. After cleaning the teats with cotton soaked in 70% alcohol, more milk was collected, and the first three to four milk jets were initially discarded, followed by the collection of 5 ml of milk in a sterile tube. The collected milk samples were then placed in an icebox and immediately transported to the laboratory for analysis. To assess the severity of mastitis, the California mastitis test (CMT) was performed as described by Abebe et al. [18], whereby 2 ml of milk from each quarter was discharged into each of the four small cups on the CMT paddle, and each cup received an equal amount of the commercial CMT reagent (DeLaval CMT Kit), after which the mixes were moved in a gently circular motion for 15 s. The details of individual milk sample collection are mentioned in Table S1–S3.

Isolation and Identification of *Staphylococcus aureus*

The isolation of *S. aureus* from milk samples was performed by inoculating one milliliter of milk into five milliliters of nutrient broth (M088, HiMedia, India) and incubating it at 37 °C for 24 h. The next day, a loopful of enriched culture was streaked onto Mannitol Salt agar (MSA) plates (M118, HiMedia, India) and incubated at 37 °C under aerobic conditions. The golden yellow colonies observed on the MSA plates, as a result of mannitol fermentation, were identified as probable *S. aureus*, while the pink colonies were disregarded. The presumptive *S. aureus* was confirmed through morphological and biochemical testing, such as Gram staining, catalase, and coagulase tests. The genotypic confirmation of *S. aureus* was performed using Polymerase Chain Reaction (PCR) targeting the coagulase *coa* and *nuc* genes. The primer details are provided in the supplementary data (Table S4) [3]. The isolates were cryopreserved in nutrient broth containing 50% (v/v) glycerol and stored at -80 °C until further processing.

Antimicrobial Susceptibility Testing

The antibiotic resistance profile of MRSA isolates was determined using an antimicrobial susceptibility test via a disc diffusion method. The assay included the use of seven antibiotics: cefoxitin (30 μg), oxacillin (1 μg), vancomycin (30 μg), erythromycin (15 μg), lomefloxacin (10 μg), tetracycline (30 μg), and chloramphenicol (30 μg) (HiMedia, India). The test was carried out on Mueller Hinton Agar (MHA) (M173, HiMedia, India) supplemented with 2.5% NaCl solution after inoculating the plates with a 0.5 McFarland standard suspension of the isolate. The plates were then incubated with antibiotic discs on agar plates for 24 h at 37 °C, after which the diameter of the zone of inhibition was determined in accordance with Clinical and Laboratory Standards Institute (CLSI) recommendations (Table S5) [19].

Genomic DNA Isolation and Genotypic Confirmation of MRSA

Genomic DNA was extracted from selected cefoxitin-resistant *S.aureus* isolates by modified procedure established in our laboratory [20]. Briefly, the isolates were first revived from frozen glycerol stock and grown on Mannitol salt agar plates at 37 °C for 24 h. The golden yellow colonies were then selected for genomic DNA extraction. A single colony was inoculated in 5 ml of Brain Heart Infusion (BHI) and incubated at 37 °C for 8 h. After adding 5 µl of ampicillin solution (50 mg/ml) to the log-phase culture and further incubation at 37 °C for 1 h, the cells were centrifuged at 5000 rpm for 5 min at 4 °C. The pellet was washed thrice with 1 ml of NaCl-EDTA buffer (30 mM NaCl, 2 mM EDTA, pH 8.0) and then treated with 200 µl of freshly prepared lysozyme solution (10 mg/ml in NaCl-EDTA buffer) for 1 h at 37 °C with intermittent shaking. After adding NaCl-EDTA and SDS solutions, the contents were incubated at 55 °C for 1 h, and conventional phenol chloroform extraction was employed for DNA isolation. The quality and quantity of the genomic DNA were evaluated using 1.0% agarose gel electrophoresis and a NanoDrop 1000 spectrophotometer. PCR amplification of the *mecA* gene, which confers methicillin resistance, was performed using appropriate primers on the presumptive verified *S. aureus* isolates for confirmation of MRSA [3].

Detection of Virulence, Enterotoxin and Antibiotic Resistance Genes

The isolated DNA was diluted in tris EDTA buffer, to a final concentration of 50 ng/µl. Monoplex PCR was performed for amplification of five virulence genes (*hlg*, *pvl*, *spa*, *coa* and *nuc*). Pyrogenic superantigen toxin *tsst 1* and enterotoxin genes (*sea* and *seg*) along with antibiotic resistance genes (*mecA*, *blaZ*) were also PCR amplified according to protocol described by Roshan et al., (2019) [3]. PCR reaction was performed on a Veriti 96-well thermal cycler (Applied Biosystems). Reaction mixture of 25 µl was prepared consisting of 1 µl (50 ng DNA) genomic DNA, DreamTaq Master Mix (K1071, Thermo Scientific, USA) containing Taq DNA Polymerase 0.25 µl (5 U/µl), DreamTaq Buffer 2.5 µl (10X), dNTP Mix 2.5 µl (2 mM), forward and reverse primer 0.25 µl (0.1 µM) and nuclease-free water added to make up final volume of 25 µl for reaction. Description of primers sequence, expected product size, and annealing temperature are summarized in supplementary material (Table S4). PCR products were analyzed on 2% agarose gel electrophoresis stained with ethidium bromide and visualized by Biorad Gel Doc XR System.

Phenotypic Evaluation of Biofilm Formation

Slime Production Assessment

The ability of *S. aureus* to generate slime was evaluated by the Congo Red Agar (CRA) method (24,275, SRL, India), with minor modifications [21]. n brief, CRA plates were prepared by adding 0.8 g of Congo red and 36 g of sucrose (94,124, SRL, India) to one litre of BHI agar. The plates were incubated at 37 °C for 24 h, followed by an additional 48 h at room temperature. The colony phenotypes were used to assess the potential for slime production. Isolates that yielded smooth, round, shiny surfaced and dry textured Bordeaux or red colonies were considered non-slime producers, while those producing black colonies with dry patterns and rough edges were categorized as slime producers [22].

Biofilm Quantification by Microtiter Plate (MTP) Method

The biofilm-forming ability of MRSA isolates was evaluated using the Microtiter Plate (MTP) method [23]. A loopful of the isolate colony grown on Mannitol Salt Agar (MSA) plates was inoculated into Trypticase Soy Broth (TSB) (24,392, SRL, India) and incubated for 24 h at 37 °C. The optical density (O.D.) of the culture was measured at 600 nm and diluted with TSB containing 1% sucrose to achieve an O.D. of 0.1 (approximately 106 cells/ml). Then, the diluted solution was added to 96-well polystyrene plates (Costar, USA) and incubated for 24 h at 37 °C. After incubation, the bacterial suspension was removed, washed three times with 200 µl of PBS, and air-dried for 15 min [24]. The samples were then fixed with 150 µl of 96% methanol for 20 min and air-dried for 60 min. The wells were stained with 200 µl of 0.2% crystal violet (28,376, SRL, India) for 15 min, and the unabsorbed stain was washed away three times with sterile distilled water. Finally, 150 µl of 96% ethanol was added to each well and incubated for 30 min to dissolve the stain. The absorbance was measured using an ELISA microplate reader at 590 nm (Tecan, Infinite M200 Pro Microplate Reader). Based on the absorbance, the bacteria were classified as non-biofilm producers ($OD \leq ODc$), weak biofilm producers ($ODc < OD \leq 2 \times ODc$), moderate biofilm producers ($2 \times ODc < OD \leq 4 \times ODc$), or strong biofilm producers ($4 \times ODc < OD$). A strong biofilm-forming strain was designated as a positive control, and sterile TSB broth without bacteria was used as a negative control [25].

Scanning Electron Microscopy Analysis of Biofilm Formation

The biofilm formation potential of *S. aureus* isolates was assessed by diluting an overnight culture 1:100 into fresh BHI broth, and adjusting the bacterial density to an optical density (OD600) of 1 at the end of incubation. A 12-well plate containing 10 mm × 10 mm glass slides in each well was filled with 2 mL of the bacterial suspension for each isolate, and the biofilm formation was allowed to occur for 48 h at 37 °C. The samples were then fixed in a 4% (v/v) glutaraldehyde solution in 0.05 M phosphate buffer (pH 7.0) for 12 h at 4 °C, followed by sequential dehydration using 30%, 50%, 70%, 90%, and 100% ethanol for 10 min per step. The dehydrated samples were sputter coated with gold using a Denton gold sputter unit for 70 s and finally mounted on a 10 kV Field Emission Scanning Electron Microscope (FE-SEM) (Apreo) [26, 27].

Detection of Biofilm and Adhesion Associated Genes

Eight genes related to MRSA biofilm formation were amplified through Polymerase Chain Reaction (PCR) techniques. These genes included *icaA, icaD* and *bap* genes, which are involved in the polysaccharide dependent and independent pathways respectively [28, 29]. Additionally, other staphylococcal adhesion genes belonging to the MSCRAMM family, including *fnbpA*, *cna*, *eno*, *ebp*, and *sasG* were also detected through PCR [11]. The aforementioned PCR protocol was implemented. The primer's sequence, expected product size, and annealing temperature are described in the supplementary material (Table S4). The PCR products were then analyzed using 2% agarose gel electrophoresis stained with ethidium bromide, and the results were visualized using the Biorad Gel Doc XR System.

Results

Prevalence of MRSA in Mastitis Milk and Antimicrobial Susceptibility Profiling

A total of 300 milk samples suspected to have mastitis were collected and analyzed. The supplementary table (Table S1–S3) includes the results of the California Mastitis Test (CMT) and sample collection details. Among the 300 samples, 135 samples were found to contain *Staphylococcus* spp on Mannitol Salt Agar (MSA), as indicated in Figure S1. The positive staphylococcal isolates were subjected to phenotypic characterization, biochemical and molecular examinations including gram staining, catalase and coagulase tests (Figures S2, S3), and detection of the *coa* gene (Figure S4). Analysis revealed that 85 out of 135 staphylococcal isolates were confirmed to be *S. aureus* (Fig. 1). Of these, 47 isolates were found to be phenotypically resistant to cefoxitin, and 46 of them were positive for the *mecA* gene, confirming them as MRSA (Figures S5, S6). The antimicrobial susceptibility profiles of these 46 MRSA isolates were determined against seven antibiotics, namely Cefoxitin (CX) (n = 46), Oxacillin (OX) (n = 42), Vancomycin (VA) (n = 12), Erythromycin (E) (n = 24), Lomefloxacin (LOM) (n = 24), Tetracycline (TE) (n = 2), and Chloramphenicol (C) (n = 0). The MRSA isolates that displayed resistance to most of the antibiotics tested were categorized as Multi-Drug Resistant (MDR), while those that showed resistance only to beta-lactam antibiotics (CX and OX) were denoted as Low Drug Resistance (LDR) isolates. The interpretation of antibiotic resistance is illustrated using a Venn diagram in Fig. 2.

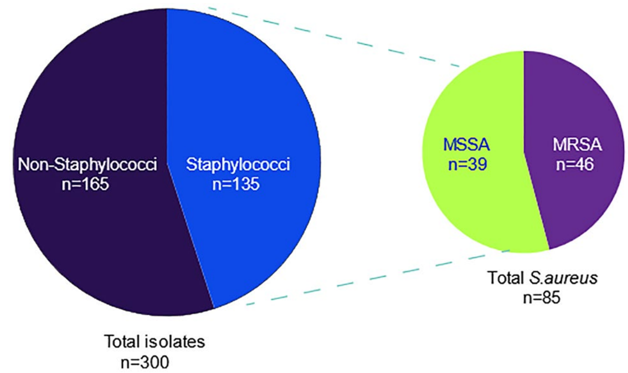

Fig. 1 Prevalence of MRSA isolates in Bovine Milk: A total 300 samples of suspected mastitis milk were included for the current study from the Haryana, India. Out of the 300 samples, 135 were found to contain *Staphylococcus* spp after being tested on mannitol salt agar. Only 47 of these isolates displayed resistance to cefoxitin and were confirmed as MRSA after testing positive for the *mecA* gene

Prevalence of Virulence Factors, Enterotoxin and Antibiotic Resistance Genes in MRSA Isolates

All 46 confirmed MRSA isolates were found to be positive for the *coa* gene, as indicated in Figures S4. Among these isolates, the thermonuclease gene (*nuc*) was detected in 35 isolates (Figures S7). Additionally, the hemolysin gene (*hlg*) and the leukocidin gene (*pvl*) were present in 36 and 14 isolates, respectively (Figures S8 and S9). Furthermore, pyrogenic toxin genes including the enterotoxin genes sea and seg were detected in 12 and 28 isolates, respectively, while the gene encoding the superantigen *tsst-1* was found in 28 isolates (Figures S10–S12). Amplification of the *spa* gene in all 46 MRSA isolates showed that 39 of them contained *spa* genes of varying lengths (Figure S13). The gene encoding antibiotic resistance, *blaZ*, was detected in 26 isolates, while *mecA* was detected in all 46 confirmed MRSA isolates (Figures S6 and S14). The frequency of genes encoding coagulase, thermonuclease, surface protein A, hemolysin, leukocidin, antibiotic resistance, and pyrogenic toxins, including enterotoxins, is presented in Fig. 3 and Table S6.

In-vitro Assessment of Biofilm Formation Potential of MRSA Isolates

The CRA assay to investigate the slime production ability of 46 MRSA isolates were employed. The results showed that 35 isolates were classified as slime producers while the remaining 11 were non-slime producers (Fig. 4a). To further assess the biofilm production capacity of these MRSA isolates, the MTP assay was employed and the OD values were recorded. Based on the observed OD values, the MRSA isolates were categorized into four groups, namely strong, moderate, weak biofilm producers, and none, with OD570

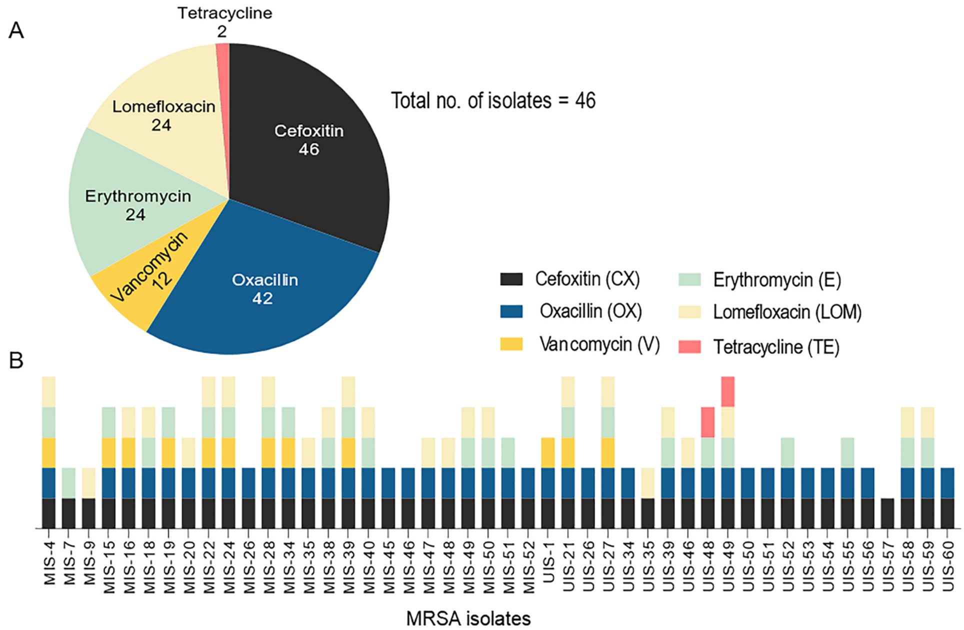

Fig. 2 Antibiotic resistance profiling: **a** Pie chart illustrate comprehensive profile of antimicrobial resistance of 46 MRSA isolates. **b** Antibiotic-susceptibility profiles of each 46 MRSA isolates determined by disc diffusion method. The colour code indicates resistance against respective antibiotic

values of ≥ 2.0, ≥ 1.0–2.0, ≥ 0.5–1.0, and ≤ 0.5, respectively (Fig. 4c). The results of the MTP assay showed that the MRSA isolates exhibited varying degrees of biofilm production capacity (Fig. 4b). Among the 46 MRSA isolates, 29 were biofilm-producing, and 17 were non-biofilm-producing isolates. Among, 29 biofilm producing isolates, 3 were classified as strong biofilm producers; 11 isolates as moderate biofilm producers and 15 isolates as weak biofilm producers (Table S7).

Scanning Electron Microscopy (SEM) Analysis and Prevalence of Genes Associated with Biofilm Formation and Surface Adhesion

The SEM was utilized to observe the architecture of biofilm formed by the *S. aureus* isolate. The morphological assessment of strong biofilm-forming and non-biofilm-forming *S. aureus* isolates, at magnifications of 15000X and 60000X (Fig. 5). Two MRSA isolates, a biofilm-producing (UIS-27) and a non-biofilm-producing (MIS-9) isolate, were selected for SEM analysis. Isolates that exhibited an OD of ≤ 0.5 did not display any structures similar to EPS, whereas those with an OD of ≥ 2.0 showed a substantial amount of EPS-like structures (Fig. 4c). The genes associated with biofilm formation, namely *icaA* and *icaD*, and surface adhesion molecule genes *bap, fnbpA*, *cna, eno, ebp*, and *sasG*, were screened (Table S1). The *icaA* and *icaD* genes were present in 40 and 43 isolates, respectively (Figure S15, Figure S16), while only one isolate tested positive for the *bap* gene. The prevalence of surface adhesion genes *ebp, fnbpA, eno, sasG*, and *cna* in 46 MRSA isolates were 43, 40, 38, 26, and 21, respectively (Figures S17–S22) (Fig. 3).

Discussion

S. aureus is a well-known etiological agent of bovine mastitis with prevalence rate of 10–40% of mastitis cases in India [3]. In the present study, we investigated the prevalence of Methicillin-Resistant *S. aureus* (MRSA) isolates in suspected bovine milk through a comprehensive analysis of virulence factors, antibiotic resistance, and biofilm-forming potential. Our investigation identified 85 *S. aureus* isolates from 300 suspected mastitis milk samples, indicating a prevalence rate of 28%. Our findings are consistent with previous reports from other countries such as China, Russia, Brazil, and Italy [30–34]. Characterization of the *S. aureus* isolates using the cefoxitin disc diffusion method and *mecA* gene-specific PCR revealed that 46 out of 85 isolates were MRSA. This represents a higher prevalence of MRSA compared to previous studies conducted in India [1, 35].

Antimicrobial therapy is the primary approach for managing bovine mastitis. However, the emergence of antibiotic resistance is diminishing the efficacy of this strategy against *S. aureus* [1]. Our investigation revealed high resistance of *S. aureus* to major beta-lactam antibiotics, with 100% of isolates resistant to cefoxitin and 91.30% resistant to oxacillin. Moderate levels of resistance were observed against erythromycin (52.17%) and lomefloxacin. Resistance to vancomycin was approximately 26.08%, while resistance to tetracycline was low (4.35%). No resistance was found against chloramphenicol. These findings are in concordance with previous reports from Malaysia, China, Egypt, and India, where high to moderate antibiotic resistance has been reported [1, 36–38]. The *coa* gene is an indispensable component of *S. aureus* and has been linked to the development of biofilms

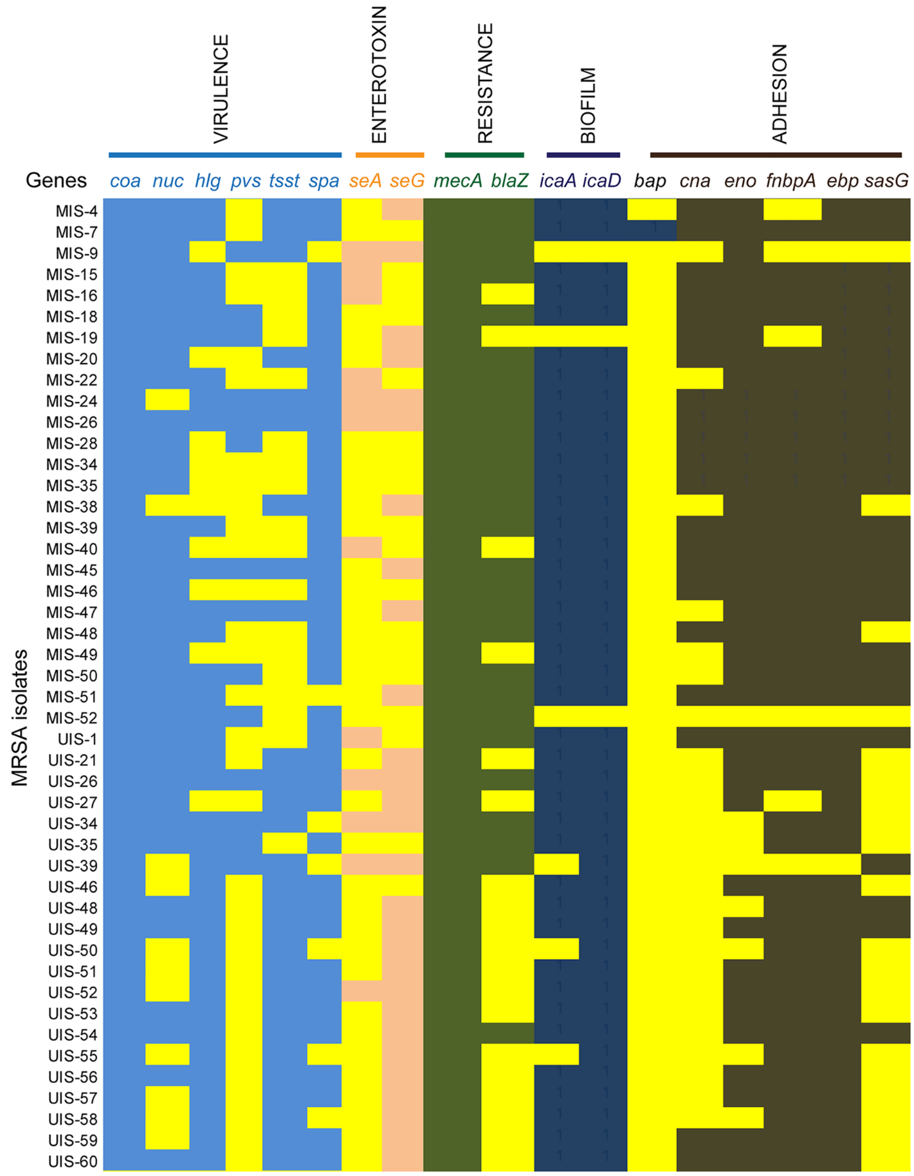

Fig. 3 Molecular profiling of Virulence determinants in MRSA isolates: Heatmap showing the prevalence of virulence, enterotoxins, antibiotic resistance, biofilm and adhesion related genes in each 46 MRSA isolates from bovine mastitis milk. The yellow colour indicates absence of genes in an isolate, respectively

[39]. Coagulase interacts with fibrinogen and triggers the creation of fibrin clots through the activation of prothrombin. The development of fibrin networks not only induces blood or plasma clotting but also shields *S. aureus* from phagocytic clearance and promotes the spread of pathogenesis in the bloodstream [40]. In this study, the *coa* gene, which encodes the coagulase protein, was discovered in MRSA isolates without any observed variation in amplicon size as previously reported [41]. The *nuc* gene, a virulence factor that contributes to pathogenicity, was detected in 76.08% of MRSA isolates [42]. The *nuc* gene encodes a thermostable nuclease that acts as a neutralizing factor in host cells, breaking down DNA and RNA. This gene also plays a part in the degradation of neutrophil extracellular traps (NETs) during local, systemic, and chronic *S. aureus* infections, enabling the pathogen to avoid NET-mediated entrapment and induce host tissue destruction [43]. The pore forming protein toxins, *PVL* and hemolysin, was present in 78.26% and 30.43% of MRSA isolates, respectively. Both of these toxins are critical in leukocytolysis, tissue necrosis, damage to the mammary epithelial lining, and pore formation in neutrophils, which increases the risk of bovine mastitis with higher mortality rates in the infected animals [44]. Finally, gene encoding antibiotic resistance, *blaZ* was present in 56.52% of isolates while *mecA* was detected in all 46 confirmed MRSA isolates, making them resistant to beta-lactam antibiotics.

S. aureus is responsible for both bovine mastitis and food-borne infections in humans [45]. Staphylococcal enterotoxins produced by *S. aureus* in food products are heat resistant and can remain present in food even after undergoing heat treatment [46]. Despite the clear role of enterotoxins in causing bovine mastitis, their persistence in

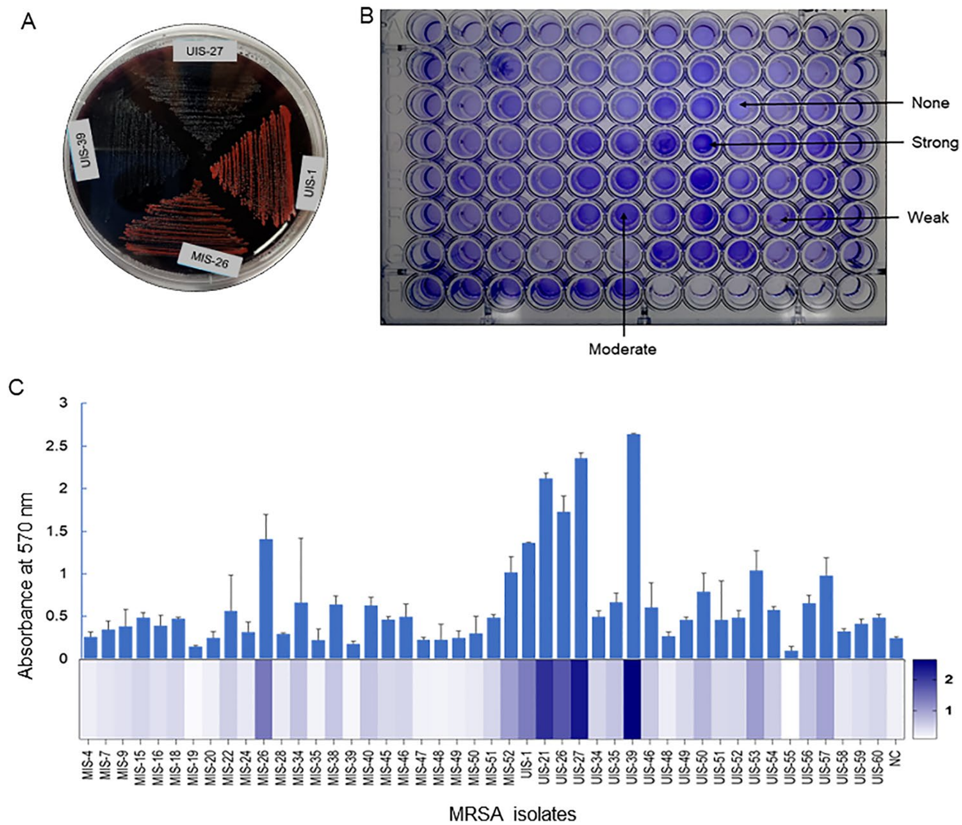

Fig. 4 Biofilm production assay: **a** CRA assay: Qualitative CRA assay were performed for biofilm production the Black colonies with a dry crystalline consistency represent isolate having biofilm producing ability and non-biofilm producing isolate were appeared with pink colonies. **b** Quantitative MTP assay: Strong, moderate and non-biofilm producers differentiated by crystal violet staining further (**c**) Biofilm formation were quantified by dissolving the crystal violet in 100% ethanol and absorbance were measured at 570 using spectrophotometer. The experiment was performed in the triplicates (n = 3) bar representing the mean value (μ)and error bar represent the standard deviation (σ). Heat map analysis (below) representing biofilm profile of 46 MRSA isolates: Strong, moderate, weak and non-biofilm producing isolates are depicted through variation in colour shades from dark blue to light blue indicating absorbance maxima to minima at 570 nm

milk post pasteurization is a significant public health concern [9]. The enterotoxin *sea* and *seg* was detected in 26.08% and 60.86% of isolates, respectively which is consistent with other studies [31]. The pyrogenic toxin gene *tsst-1* was found in 60.8% of isolates, which is higher than previously reported prevalence rates [47]. TSST-1 triggers excessive cytokine production by activating T cells indiscriminately, leading to multi-organ dysfunction and death [44, 48]. The *spa* gene encodes Protein A, which inhibits opsonization and phagocytosis to evade host immune responses [3]. In present study, *spa* gene was detected in 84.78% of MRSA isolates, with the gene length observed to vary between 200 and 500 base pairs (bp), consistent with previous research that identified amplicons ranging from 140 to 450 bp [3, 49].

According to the qualitative CRA biofilm assay, it was found that 76% of the MRSA isolates generated black colonies, which indicates the presence of PIA and slime production, both of which are biofilm structural components. The more reliable, sensitive, specific and precise MTP assay was used to analyze the biofilm production of confirmed MRSA isolates, and it showed that 63% of these isolates were biofilm producers [50]. This result is in contrast to earlier studies that reported a higher percentage of biofilm-forming isolates [30, 38]. Out of the total MRSA isolates, 10% were strong biofilm producers, 37.9% were moderate biofilm producers, and 51.7% were weak biofilm producers. Analysis of the biofilm architecture revealed that the biofilm-forming isolate (UIS-27) was structured in multiple layers with large cell clusters and a polysaccharide-based elastic organization, whereas the non-biofilm-forming isolate (MIS-9) did not show uniform attachment and was heterogeneous. [51]. The *ica* operon regulates the production of PIA-dependent biofilm in *S. aureus* [13]. Our research found that the isolates had a high prevalence of the *icaA* and *icaD* genes, with 86.9% and 93.4% of the isolates testing positive for these genes, respectively. These findings are consistent with previous reports, which have also documented high prevalence rates ranging from 90 to 100% for these genes [1,

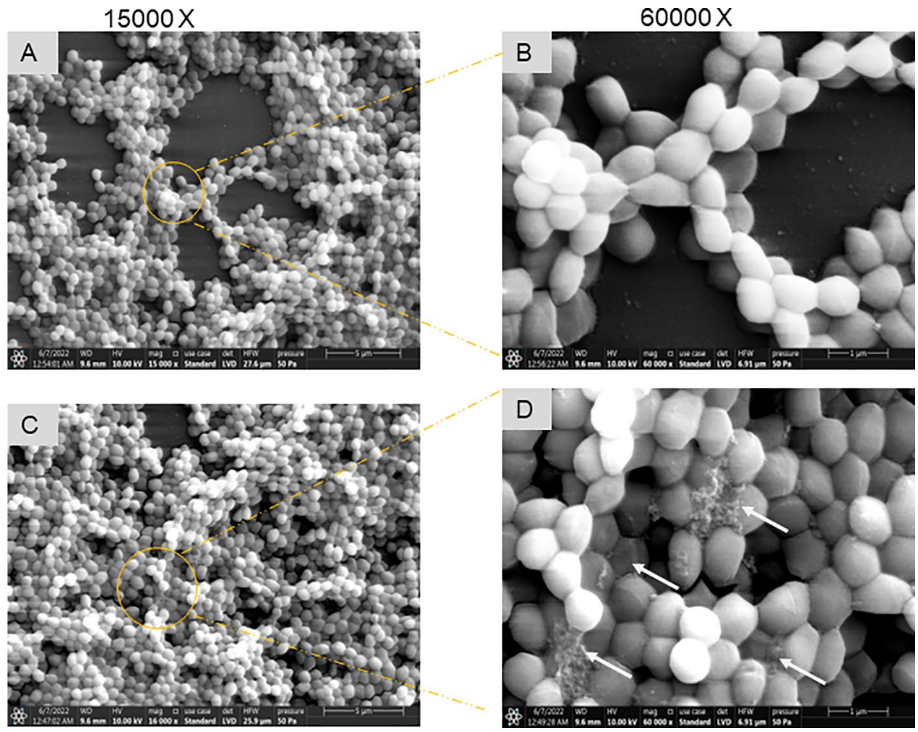

Fig. 5 SEM analysis of non-biofilm and biofilm forming MRSA: Image **a** and **b**: A non-biofilm forming isolate (MIS-9), Image **c** and **d**: A biofilm producing isolate (UIS-27). Image was captured at 15000X and 60000X to evaluate biofilm production in *S. aureus* (MRSA). White arrows indicate the presence of EPS matrix, fundamental characteristic of the biofilm. The circle in SEM image represents the magnified images of bacterial cells

24, 30]. Additionally, only one isolate tested positive for the bap gene, which is consistent with recent studies examining bovine mastitis and bovine mammary epithelial cells [1, 30].

Staphylococcal MSCRAMMs play a vital role in the initial attachment of *S. aureus* to host cells and also in the adhesion to medical equipment [13, 52]. The FnbpA protein, which is involved in the adherence and invasion of bovine mammary epithelial cells, was found in 86.9% of the isolates, consistent with previous studies [1]. Additionally, the *eno* gene product, α-enolase, was detected in 82.6% of the isolates, which aids in binding *S. aureus* to laminin and adhesion to the extracellular matrix [53]. The *ebps* gene-encoded Ebps protein, a cell wall integral membrane protein, was present in 93.4% of the isolates, consistent with previous research on raw milk and bovine mastitis in India and China [54], where a high prevalence rate of *ebps* gene of 90% and 80.8%, respectively, have been reported [41, 55]. The prevalence rates of the *cna* and *SasG* genes, responsible for collagen adhesion and surface protein G, respectively, were reported to be 45.6% and 56.5% in our study, which contrasts with previous studies reporting higher prevalence rates of these genes in *S. aureus* from bovine mastitis. [1, 41].

The ability of pathogenic staphylococci to create biofilms is a significant factor in their capacity to colonize mammary alveolar epithelium and establish long-term persistence, acting as a source for bacterial spread through milk secretion and contamination of healthy animals [13]. Bacterial cells that exist within biofilms have a low metabolic rate and reproduce quickly, leading to an increase in antibiotic tolerance and resistance even at higher than minimum inhibitory concentration levels [12]. Several MRSA isolates that produce biofilms also possess virulence characteristics and antibiotic resistance genes, which could lead to persistent and hard-to-treat chronic infections. In this study, the distribution of various resistance phenotypes was analyzed among 46 MRSA isolates with different biofilm-forming abilities, indicating that all three MRSA isolates with a strong ability to form biofilms were MDR, while the 11 MRSA isolates with moderate biofilm-forming ability were all LDR. Of the 15 MRSA isolates with weak biofilm-forming ability, eight were MDR and seven were LDR, accounting for 53% and 47% respectively. Among the non-biofilm producing strains, ten (60%) were MDR and seven (41.7%) were LDR. The formation of biofilms within glandular epithelium leads to inflammation and a decrease in milk production, creating a favorable environment for bacterial growth. Therefore, the ability to form biofilms may be a crucial indicator of a pathogen that necessitates effective elimination from dairy animals. As such, identifying the genes responsible for biofilm formation and patterns of pathogen and antibiotic resistance is critical in preventing colonization of mammary epithelial cells and the formation of biofilms by the pathogen.

Conclusion

The present study provides a comprehensive view of virulence factors, antibiotic resistance profile, and biofilm-forming potential of MRSA isolates *in-vitro*. 300 bovine milk samples tested, 15% were MRSA, with the highest resistance observed against cefoxitin (100%) and oxacillin (91.30%), and moderate resistance (52.17%) against erythromycin and lomefloxacin. MRSA strains with the ability to form biofilms, 40% belonged to the MDR phenotype, demonstrating resistance to multiple antibiotics including Cefoxitin, Oxacillin, Vancomycin, Erythromycin, and Lomefloxacin. However, no correlation was found between the resistance profile and the ability to form biofilms. In current study most of these samples were screened from chronic mastitis cases and evaluated *in-vitro* for biofilm forming potential. Our study had three major limitations:

A. The cultures were not studied and evaluated *in-vivo* in live infected animals to support our hypothesis.
B. Biofilm-forming potential of *S. aureus* in experimental models was not evaluated to investigate the strategies employed by *S. aureus* to survive within mammary gland that enables them resist anti-microbials treatment.
C. The relationship between antimicrobial resistance and biofilm formation in *S. aureus* is not properly correlated, and this concept needs further investigation.

Therefore, it is crucial to establish an efficient diagnostic procedure for detecting biofilm.forming MRSA strains in chronic bovine mastitis cases, and to develop new methods to prevent biofilm formation. Several potential approaches exist to achieve this, including developing inhibitory molecules that can block biofilm formation, creating vaccines targeting capsular polysaccharides, utilizing bio-engineered phages, and using plant-derived crude extracts and fractions as therapeutic agents for treating and eliminating biofilms.

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Author Contributions IS: Methodology, Investigation, Writing-Original Draft. MR: Methodology, Conceptualization, Investigation, Writing-Original Draft. AV: Conceptualization, Visualization, Formal Analysis, Writing—Review & Editing. DG and SR: Formal analysis, Investigation. MB and CR: Recourses. SD: Supervision. Conceptualization, Investigation, Funding acquisition, Writing-Review & Editing.

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Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

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